UNITED ARAB EMIRATES MINISTRY OF EDUCATION
الإمارات العربية المتحدة وزارة التربية والتعليم

Al Khalidiya Boys School – Cycle 3

School code 10048 | Dibba Al Hisn | Sharjah Eastern

Grade 9 Physics Final Exam Practice

Term 1 2026–2027 • Set C • With Answers

100 marks | 150 minutes | Calculator allowed

Answer all 20 questions. Choose one option for each MCQ. Show working, units and labelled diagrams for Q17–20. Use g = 9.8 m/s² when needed.

Section A Multiple Choice 60 marks

1. Why should a scientific investigation be described clearly enough for other scientists to repeat? [4 marks]

A. To eliminate the need for measurements
B. To check whether the results are reproducible
C. To guarantee that the hypothesis is correct
D. To turn an opinion into a scientific law

Reference: p. 8, Check Your Progress 3 | **Answer B**

2. What is an important limitation of a scientific model? [4 marks]

A. It cannot be modified after new evidence
B. It represents selected features rather than every detail of a real system
C. It must always be a physical miniature
D. It has to predict every possible event

Reference: pp. 6–7 | **Answer B**

3. Which statement about scientific theories is correct? [4 marks]

A. They explain observations and are supported by evidence
B. They automatically turn into laws after many experiments
C. They are less useful than guesses
D. They cannot be tested

Reference: p. 7 | **Answer A**

4. A time is 250 µs. What is this time in seconds? [4 marks]

A. 2.50×10^{-2} s
B. 2.50×10^{-9} s
C. 2.50×10^{-4} s
D. 2.50×10^{4} s

Reference: pp. 10–11 | **Answer C**

5. How many significant digits are in 0.00450 m? [4 marks]

A. 2
B. 4
C. 3
D. 5

Reference: pp. 11–12 | **Answer C**

6. Which action mainly checks the accuracy of a balance? [4 marks]

A. Measure more quickly
B. Add extra digits to its display
C. Repeat a reading without a reference mass
D. Compare its reading with a known standard mass

Reference: p. 14 | **Answer D**

7. A mass–volume graph has a nonzero vertical intercept. What can the intercept represent? [4 marks]

A. A changing density at every point
B. The gravitational acceleration
C. The mass of an empty container
D. The volume of an empty container

Reference: p. 22, Check Your Progress 20 | **Answer C**

8. Two runners start together and move at constant speeds. At one instant runner A is at the finish and runner B is three-fourths of the way there. What follows? [2 marks]

A. A must have stopped halfway
B. A has travelled farther in the same elapsed time
C. Their speeds must be equal
D. B has travelled for a longer time

Reference: p. 31, Check Your Progress 5 | **Answer B**

9. A runner travels 100 m north. Which part makes the displacement a vector description? [4 marks]

A. The stated direction north
B. The unit metre
C. The number 100
D. The runner's mass

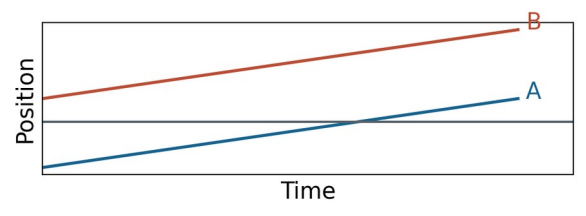
Reference: p. 33 | **Answer A**

10. Ari moves 2.6 m every 2.0 s along a school corridor. Which graph represents his motion from the cafeteria? [4 marks]

A. A horizontal position–time line
B. A line with slope 5.2 m/s
C. A curve whose slope continually increases
D. A straight line with slope 1.3 m/s

Reference: p. 39, Practice Problem 14 | **Answer D**

11. On the graph, which pedestrian begins west of the origin? [4 marks]



A. Neither
B. A
C. B
D. Both

Reference: p. 39, Practice Problem 13 | **Answer B**

12. In which case can average and instantaneous velocities differ? [4 marks]

A. A car changes its velocity during the measured interval
B. An object has exactly constant velocity
C. A car maintains the same velocity throughout
D. A stationary object stays at rest

Reference: pp. 43–44 | **Answer A**

13. A ball rises vertically after being thrown upward. Neglect air resistance. Which statement is correct while it rises? [4 marks]

A. Its downward acceleration becomes zero
B. Its speed increases throughout the rise
C. Its upward velocity decreases because acceleration is downward
D. Its acceleration is upward

Reference: pp. 56; 71–74 | **Answer C**

14. A car accelerates from rest at 5.5 m/s² north. How long does it take to reach 28 m/s north? [4 marks]

A. 5.1 s
B. 33.5 s
C. 154 s
D. 0.20 s

Reference: p. 65, Practice Problem 18 | **Answer A**

15. A skateboarder starts uphill at 1.75 m/s and slows at 0.20 m/s². How long before reversing direction? [4 marks]

A. 1.55 s
B. 17.5 s
C. 0.35 s
D. 8.75 s

Reference: p. 68, Practice Problem 23 | **Answer D**

16. A ball is in free fall near Earth. Taking upward as positive, its acceleration is approximately: [2 marks]

A. −9.8 m/s
B. 0 m/s² at every point
C. +9.8 m/s² only while rising
D. −9.8 m/s²

Reference: pp. 71–74 | **Answer D**

Set C Section B Free Response

17. In the spring investigation, added mass is changed and the spring length is measured. Three measurements of one spring are (14.6 ± 0.2) cm, (14.8 ± 0.3) cm and (14.0 ± 0.1) cm. [10 marks]

Reference: pp. 11–14; pp. 17–18, Spring Investigation

(a) Identify the independent and dependent variables and the axis for each. [3]

Answer Independent mass on horizontal axis [1]; dependent spring length on vertical axis [1]; correct units on both axes, e.g. kg and m or labelled g and cm [1].

(b) Name one quantity that should be controlled. Explain why. [2]

Answer Use the same spring [1], so changes in length are due to added mass rather than different spring properties [1].

(c) Convert (14.6 ± 0.2) cm into metres. [2]

Answer (0.146 ± 0.002) m [1 for value, 1 for uncertainty].

(d) Explain precision versus accuracy using the results if the accepted length is 14.8 cm. [3]

Answer Precision concerns spread/uncertainty [1], so ±0.1 cm is most precise [1]. Accuracy concerns agreement with accepted value, so 14.8 cm is most accurate [1].

18. An object moves from x = 65 cm to x = 37 cm in 2.3 s without reversing direction. Separately, a car's velocity changes from 3.0 m/s west to 4.5 m/s east in 2.5 s. Take east as positive for the car. [10 marks]

Reference: p. 45, Practice Problem 28; p. 61, Practice Problem 8

(a) Find the object's displacement in metres. [2]

Answer Δx = 37 − 65 = −28 cm [1] = −0.28 m [1].

(b) Find the object's average velocity and average speed. [3]

Answer $v_{average}$ = −0.28/2.3 = −0.122 m/s [2: method and result]; speed = +0.122 m/s [1].

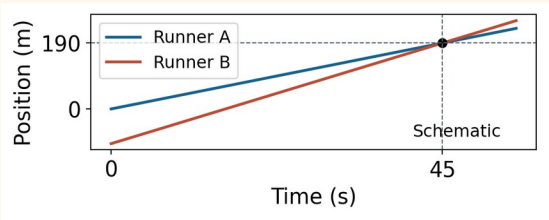
(c) Calculate the car's velocity change. [2]

Answer Δv = +4.5 − (−3.0) [1] = +7.5 m/s [1].

(d) Calculate its average acceleration and explain the positive sign. [3]

Answer a = 7.5/2.5 [1] = +3.0 m/s² [1]; positive means eastward acceleration, even while the car initially moves west [1].

19. The textbook graph for two runners shows their position lines intersecting at approximately t = 45 s and x = 190 m. Runner B's line is steeper and increasing; runner A's line is also increasing. Use the labelled sketch. [10 marks]



Reference: p. 40, Example Problem 2; pp. 37–39

(a) When and where does runner B pass runner A? [2]

Answer About 45 s [1], at about +190 m from the origin [1].

(b) Which runner has greater velocity? Explain using the graph. [2]

Answer Runner B [1], because its positive position–time slope is greater [1].

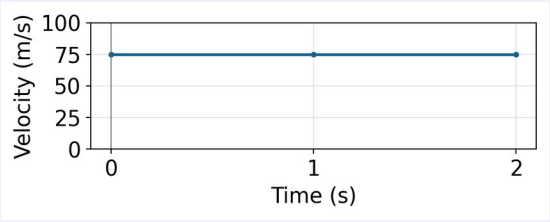
(c) Do equal positions prove equal velocities? Explain. [2]

Answer No [1]; position is the graph value, velocity is its slope, and their slopes differ [1].

(d) Does the intersection necessarily mean a collision? What does a straight increasing position line mean? [4]

Answer No [1]; they may run in different lanes [1]. Straight line means constant velocity [1]; increasing position means motion in the positive direction [1].

20. An airplane moves at a constant 75 m/s in the positive direction. The graph is a representation of the textbook example over the first 2.0 s. [10 marks]



Reference: p. 66, Example Problem 3; pp. 57–59

(a) Determine its acceleration from the graph. [2]

Answer a = slope [1] = 0 m/s² [1].

(b) Use graph area to calculate displacement in the first 1.0 s and first 2.0 s. [3]

Answer Rectangle area vt [1]; 75 m in 1.0 s [1]; 150 m in 2.0 s [1].

(c) Draw the corresponding displacement–time graph for 0–2.0 s, taking initial position as zero. [3]

Answer Correct labelled axes [1]; straight line through origin [1]; through (1,75) and (2,150) [1].

(d) Explain how slope and area on the given velocity graph represent different quantities. [2]

Answer Slope gives acceleration [1]; signed area gives displacement [1].

